

PhageLytic-ESM: Bacteriophage Lysin Catalytic Domain Classification via Deep Semantic Embeddings

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Abstract—The crisis of Antimicrobial Resistance drives the search for alternative therapeutics, notably bacteriophage endolysins. A critical bottleneck in deploying these lysins is the accurate classification of their catalytic domains (Amidases, Glycosylases, Peptidases). Traditional bioinformatics relies on classical sequence alignment and 1-dimensional biophysical feature extraction, which frequently fail to capture the 3-dimensional structural conservation required for precise functional annotation. In this paper, we introduce PhageLytic-ESM, a supervised deep learning architecture that bypasses classical feature engineering by leveraging a 150-million-parameter Protein Language Model (ESM-2). By extracting 640-dimensional latent embeddings, we map raw amino acid sequences into a deep semantic space that implicitly captures biological topology. Evaluated on a balanced dataset curated from PhaLP 2.0, our Tuned Multi-Layer Perceptron (MLP) achieves 98.61% cross-validated accuracy. Crucially, we deploy a strict cluster-aware StratifiedGroupKFold validation protocol. While sequence identity analysis confirms high homology (mean 78.89%) within the dataset, our splitting methodology guarantees zero evolutionary leakage, proving the model learns generalizable structural logic rather than memorizing local sequence artifacts. Finally, a simulated deployment using natural class prevalence ratios from the Global Cold Seep Virome (GCSV) projects greater than 95% precision across all classes, validating the architecture’s viability for high-throughput, extreme-environment metagenomic pipelines, such as deep-sea bioprospecting in areas like Philippine Rise.

Index Terms—Bacteriophage Lysins, Protein Language Models, Sequence Alignment, Out-of-Distribution Validation

I. INTRODUCTION

The escalating crisis of Antimicrobial Resistance (AMR) has accelerated the search for alternative therapeutics, with bacteriophage endolysins (lysins) emerging as highly promising candidates [1]. Lysins are phage-encoded peptidoglycan hydrolases that rapidly lyse bacterial cell walls. To destroy the cell wall, lysins deploy highly specific mechanical functions, broadly categorized into three primary catalytic domains based on their exact cleavage sites: Glycosylases cleave the sugar backbone, Peptidases sever the cross-linking peptide bridges, and Amidases cleave the connecting amide bonds between the sugars and peptides [2]. Accurately classifying these

specific domain architectures is a critical bottleneck in both computational biology and wet-lab target synthesis pipelines.

Traditional bioinformatic approaches rely heavily on classical sequence alignment. While highly effective for closely related evolutionary cousins [3], alignment tools are fundamentally limited by their dependence on 1-dimensional (1D) character matching. As target sequences mutate, classical algorithms struggle to capture the underlying 3-dimensional (3D) structural conservation that dictates catalytic function. Furthermore, traditional machine learning models utilizing basic biophysical properties (e.g., molecular weight, isoelectric point, pseudo-amino acid composition) consistently plateau in accuracy because they lack deep spatial context [4].

To bridge this gap, we introduce PhageLytic-ESM, a supervised deep learning architecture that bypasses classical 1D feature engineering. By leveraging a 150-million-parameter Protein Language Model (ESM-2) [5], we extract dense, 640-dimensional latent embeddings that capture the inherent biological “grammar” and 3D geometric topology of lysin sequences.

The primary contribution of this work is threefold:

- We demonstrate that deep semantic embeddings dramatically outperform classical biophysical features in catalytic domain classification.
- We deploy a rigorous, homology-aware StratifiedGroupKFold validation protocol. By strictly vaulting entire evolutionary clusters (defined via standard homology thresholds [6]), we can guarantee that our Multilayer Perceptron (MLP) architecture achieves its cross-validated accuracy by learning generalizable structural logic, rather than memorizing local sequence artifacts [7].
- We project our model’s performance against natural class prevalence ratios derived from global deep-sea viromes, demonstrating its immediate viability for extreme-environment biodiscovery (e.g., the Philippine Rise).

II. RELATED WORK

A. Classical Sequence Alignment in Bioinformatics

For decades, local and global sequence alignment algorithms have served as the industry standard for functional annotation [3], [8]. These tools calculate statistical similarity based on exact amino acid matches and gap penalties. While highly reliable within homologous regimes, they inherently struggle to identify remote homologs where sequences have diverged significantly but 3D catalytic structures remain conserved. Our sequence identity analysis confirms that while standard local alignment operates effectively within our dataset’s high-homology clusters, relying solely on alignment limits the scalable discovery of highly divergent, novel architectures.

B. Machine Learning and Biophysical Feature Engineering

To overcome the limitations of pure sequence matching, researchers have historically employed classical machine learning algorithms trained on extracted 1D biophysical properties [4]. Features computed from primary sequences, such as molecular weight, isoelectric point (pI), instability indices, and Grand Average of Hydropathy (GRAVY) scores, provide a bulk summary of a protein’s physical and chemical nature. However, models relying strictly on these parameters frequently plateau in predictive power. Because 1D human-engineered features discard the precise topological arrangement of the amino acids, they act as blunt instruments, incapable of mapping the complex, non-linear boundaries of multi-domain catalytic sites.

C. Protein Language Models (pLMs) and Deep Semantics

The introduction of transformer-based Large Language Models to structural biology has fundamentally shifted computational protein analysis. Models such as Evolutionary Scale Modeling (ESM-2), trained on hundreds of millions of unannotated protein sequences, learn the evolutionary and biophysical constraints of amino acid arrangements [5]. By treating sequences as a biological language, pLMs generate high-dimensional latent representations (embeddings) that implicitly encode 3D secondary and tertiary structures [9]. This paper extends this paradigm by applying deep semantic embeddings specifically to the automated, high-precision classification of phage lysin catalytic domains, isolating the true structure from evolutionary noise.

III. DATASET DESCRIPTION

Protein sequences were sourced from PhaLP 2.0 (v2025_06) [10], [11], a curated and metagenomically extended database of bacteriophage lytic proteins hosted at <https://phalp.ugent.be>. PhaLP 2.0 integrates lysin entries from UniProt with over 743,000 sequences mined from the EnVhogDB viral protein database using the SUBLYME embedding-based classifier, and organizes entries into cluster-centered representations with per-entry metadata including protein identifiers, amino acid sequences, cluster

assignments, conserved domain annotations, and enzymatic activity classifications.

Three enzyme classes were queried via the PhaLP 2.0 Custom Tables interface. Amidases (EC 3.5.1.28) and Glycosylases/Lysozymes (EC 3.2.1.17) were retrieved by Enzyme Commission number, while Peptidases were retrieved by domain filter (Peptidase_M15 and Peptidase_M23 metalloendopeptidase domains). Each record included the protein ID, protein name, amino acid sequence, cluster name, and sequence length. The merged raw dataset contained 77,105 sequences distributed across 2,572 Amidases, 5,077 Glycosylases, and 69,456 Peptidases.

Table I summarizes the dataset composition at each preprocessing stage.

TABLE I
DATASET COMPOSITION AFTER EACH PREPROCESSING STAGE

Stage	Amid.	Glyc.	Pept.	Total
Raw query	2,572	5,077	69,456	77,105
After cleaning	1,591	3,293	65,570	70,454
After balancing	1,591	1,591	1,591	4,773

For the real-world deployment projection, class prevalence ratios were derived from the Global Cold Seep Virome (GCSV) [12], a catalog of 193,465 species-level viral operational taxonomic units reconstructed from 314 metagenomes spanning 18 cold seep sites worldwide. These ratios are used to contextualize our evaluation and report deployment-aware performance metrics relevant to deep-sea settings (e.g., the Philippine Rise), rather than laboratory-only accuracy. Specifically, the 5,953 high-confidence lysins reported in the GCSV (Table S13 of [12]) were mapped to the three target classes by InterPro domain annotation, yielding natural prevalence ratios of 30.71% Amidase, 31.42% Glycosylase, and 37.87% Peptidase.

IV. METHODOLOGY

A. Data Collection and Curation

The raw dataset of 77,105 bacteriophage lysin sequences across three enzyme classes was collected from PhaLP 2.0 as described in Section III. The dataset exhibited a severe class imbalance: 69,456 Peptidases (90.08%), 5,077 Glycosylases (6.58%), and 2,572 Amidases (3.34%), necessitating the balancing strategy detailed below.

B. Preprocessing

Several cleaning steps were applied sequentially. Rows with missing protein sequences were removed, all sequences were standardized to uppercase, and exact-duplicate sequences were dropped. Sequences were then filtered to retain only those between 50 and 1,000 amino acids in length, which removed truncated fragments and abnormally long multi-domain proteins that could introduce noise. An additional pass removed sequences containing non-standard amino acid characters. After all cleaning steps, 70,454 sequences remained. To eliminate

class imbalance, all classes were randomly downsampled to match the smallest class (1,591 Amidases), producing a balanced dataset of 4,773 sequences. The random seed was fixed at 42 throughout. The full preprocessing pipeline is summarized in Table I.

C. Classical Feature Extraction

Four biophysical properties were computed per sequence using BioPython’s ProteinAnalysis module [13]: molecular weight, isoelectric point (pI), grand average of hydropathicity (GRAVY), and instability index. These properties were used both for exploratory analysis and as a baseline feature set.

D. ESM-2 Embedding Generation

To capture sequence-level structural information beyond what scalar biophysical features can represent, each protein sequence was passed through ESM-2 (esm2_t30_150M_UR50D), a 150-million-parameter protein language model pretrained on the UniRef50 database [5]. The model was used as a frozen feature extractor: no weights were updated during this step. For each sequence, the final hidden state was mean-pooled across the sequence length dimension, producing a single 640-dimensional embedding vector. The resulting feature matrix had shape $(4,773 \times 640)$. The 150M variant was selected as a practical tradeoff between representation quality and compute cost.

E. Cluster-Aware Data Splitting

Protein sequence datasets contain evolutionary relatives that share high sequence similarity. A naive random split risks placing homologous sequences in both training and test sets, inflating accuracy through what amounts to partial memorization. To prevent this, StratifiedGroupKFold ($K = 5$) from scikit-learn [14] was used, with the PhaLP 2.0 cluster name as the grouping variable. This guarantees that all sequences from a given evolutionary cluster appear exclusively in either the training or test set, while preserving approximate class balance within each fold. A leakage audit confirmed zero cluster overlap between training and test partitions across all folds. A naive `train_test_split` was also run for comparison to quantify the effect of homology leakage.

F. Classification Models

Six classifiers were trained on the ESM-2 embeddings: Multi-Layer Perceptron (MLP), XGBoost, Random Forest, Logistic Regression, K -Nearest Neighbors (KNN), and Gaussian Naive Bayes. All models used default hyperparameters in the initial comparison and were evaluated using 5-fold Stratified-GroupKFold cross-validation. Performance was measured by mean accuracy ($\pm$ standard deviation) across folds, along with per-class precision, recall, and F1-score from the final held-out fold.

G. Feature Ablation

To test whether classical biophysical features add information beyond what the ESM-2 embeddings already encode, a hybrid feature set was constructed by concatenating the 4 BioPython features with the 640 ESM-2 dimensions, yielding 644 features per sequence. All six classifiers were retrained and re-evaluated under the same 5-fold cross-validation protocol.

H. Hyperparameter Tuning

A randomized search (RandomizedSearchCV, 20 configurations) was run on the MLP to determine whether architectural tuning could separate it from the other converging classifiers. The search space is detailed in Table II. Cross-validation used the same StratifiedGroupKFold ($K = 5$) splitter.

TABLE II
MLP HYPERPARAMETER SEARCH SPACE

Parameter	Search Range
Hidden layers	(64,), (100,), (128,), (256,), (100, 50), (128, 64), (256, 128)
Activation	ReLU, tanh
α (L2)	$10^{-5} - 10^{-1}$ (log-uniform)
Learning rate	$10^{-4} - 10^{-2}$ (log-uniform)

I. Sequence Identity Analysis

Smith-Waterman local alignment [8] was used to compute maximum pairwise sequence identity between sampled test and training sequences (50 test vs. 500 training) to characterize the homology regime of the dataset and assess the degree of sequence divergence between partitions.

J. Evaluation

Final model performance was assessed through confusion matrices, per-class precision/recall/F1, and learning curves from the tuned MLP. Early stopping (validation fraction = 0.1) halted training when validation performance plateaued. All six default classifiers and the tuned MLP were retrained and evaluated on the canonical held-out fold (the last fold of the StratifiedGroupKFold split) to produce a direct head-to-head ranking under identical conditions. A real-world deployment projection was computed using the GCSV class prevalence ratios to estimate precision under realistic environmental class distributions.

V. EXPERIMENTS AND RESULTS

A. Classical Baseline

To establish a lower bound on classification performance using conventional feature engineering, all six classifiers were trained on the four BioPython features (molecular weight, pI, GRAVY, instability index) under a standard train/test split. The best classical accuracy was 82.83%, achieved by XGBoost [15]. This ceiling reflects the near-complete distributional overlap of pI, GRAVY, and instability across the three enzyme classes observed during exploratory analysis, confirming that

scalar biophysical properties alone cannot resolve catalytic domain boundaries.

B. ESM-2 Embedding Performance

Replacing the four biophysical features with the 640-dimensional ESM-2 embeddings produced an immediate and substantial improvement across all classifiers under 5-fold StratifiedGroupKFold cross-validation. Table III reports the full comparison across feature sets. The convergence of the top four models within a narrow accuracy band is a central finding: Logistic Regression, a linear classifier, matches the MLP, indicating that the ESM-2 embedding space is nearly linearly separable across the three enzyme classes. This demonstrates that classification performance is driven primarily by embedding quality rather than classifier complexity.

TABLE III
CROSS-VALIDATED ACCURACY (%) UNDER CLUSTER-AWARE SPLITTING

Model	ESM-Only (640d)	Hybrid (644d)
Logistic Regression	98.26 $\pm$ 0.72	98.28 $\pm$ 0.77
MLP	98.18 $\pm$ 0.74	98.39 $\pm$ 0.70
XGBoost	97.84 $\pm$ 0.41	97.86 $\pm$ 0.41
Random Forest	97.49 $\pm$ 0.68	97.46 $\pm$ 0.86
KNN	96.94 $\pm$ 1.51	96.98 $\pm$ 1.39
Naive Bayes	90.09 $\pm$ 1.49	90.03 $\pm$ 1.48
Tuned MLP	98.61 $\pm$ 0.41	—
Best classical baseline (XGBoost, BioPython features): 82.83%.		

C. Effect of Cluster-Aware Splitting

A naive `train_test_split` estimated MLP accuracy at 99.06%, compared to 98.18% under cluster-aware cross-validation, which is a 0.88 percentage point gap. This difference reflects homology leakage in the naive approach, where evolutionarily related sequences appear in both training and test partitions. The result validates the necessity of Stratified-GroupKFold for biological sequence data.

D. Feature Ablation

Cross-validated comparison of the ESM-only feature set (640 dimensions) against the hybrid feature set (644 dimensions) showed no statistically significant change for any model, as shown in Table III. All accuracy deltas fell within $\pm 0.3\%$, well inside one standard deviation. The biophysical features were therefore deemed fully redundant, consistent with the interpretation that the protein language model’s latent representations already encode the physicochemical properties that classical features attempt to capture.

E. Hyperparameter Tuning

The tuned MLP (two hidden layers of 256 and 128 units, $\alpha \approx 0.0012$, learning rate ≈ 0.00045 , ReLU activation) achieved 98.61% $\pm$ 0.41% under the same 5-fold cross-validation protocol (Table III). This represents a modest gain over the default MLP with notably tighter fold-to-fold variance, confirming that architectural tuning provides marginal

improvement but does not fundamentally shift the performance ceiling set by the embeddings.

On the canonical held-out fold, the tuned MLP achieved 98.01% test accuracy. Early stopping halted training at epoch 23, with a final training loss of 0.0031 and peak validation accuracy of 99.74%, indicating clean convergence without overfitting.

F. Error Analysis

Table IV presents the confusion matrix from the tuned MLP on the canonical test fold (954 samples). Of the 19 total misclassifications, 11 involved Peptidases predicted as Glycosylases and 4 involved Glycosylases predicted as Peptidases. This confusion pattern is consistent with the known biological reality that these two catalytic classes operate on overlapping regions of the peptidoglycan matrix and can exhibit multi-domain or hybrid architectures that blur functional boundaries [2].

TABLE IV
CONFUSION MATRIX AND PER-CLASS METRICS — TUNED MLP ON CANONICAL TEST FOLD ($n = 954$)

Actual	Predicted		
	Amidase	Glycosylase	Peptidase
Amidase	317	1	0
Glycosylase	3	311	4
Peptidase	0	11	307

Class	Precision	Recall	F1-Score
Amidase	0.9906	0.9969	0.9937
Glycosylase	0.9628	0.9780	0.9704
Peptidase	0.9871	0.9654	0.9762

G. Sequence Identity Analysis

Smith-Waterman local alignment [8] between 50 sampled test sequences and 500 training sequences showed a mean maximum identity of 78.89% (min: 72.41%, max: 92.50%). None of the sampled test sequences fell below the 40% “twilight zone” threshold [16] where classical alignment tools become unreliable. This confirms that the dataset’s sequences remain within the homologous regime, placing the methodological burden on cluster-aware splitting rather than sequence divergence as the primary safeguard against data leakage [6], [7]. The full distribution of maximum pairwise identities is shown in Fig. 1.

H. Real-World Deployment Projection

Using class prevalence ratios derived from the GCSV metagenome [12] (30.7% Amidase, 31.4% Glycosylase, 37.9% Peptidase), the tuned MLP projects 99.04% precision for Amidase, 95.62% for Glycosylase, and 98.93% for Peptidase under a simulated deployment of 10,000 sequences. The slight precision drop for Glycosylase is consistent with the confusion matrix analysis and reflects the inherent structural overlap between Glycosylase and Peptidase catalytic domains

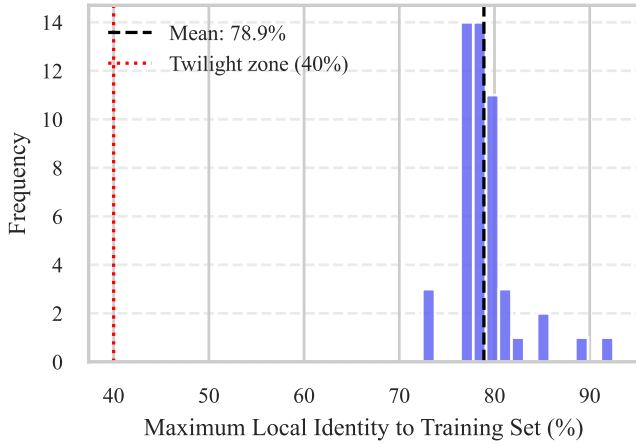


Fig. 1. Distribution of maximum pairwise sequence identity between test and training sequences (Smith-Waterman, $n = 50$ test vs. 500 training). The dashed line marks the mean (78.9%). No test sequence fell below the 40% twilight zone threshold.

in environmental samples [2], [12]. The projected confusion matrix under GCSV prevalence is shown in Fig. 2.

Projected Deployment ($n = 10,000$, GCSV Prevalence)

True Class	Predicted Class		
	Amidase	Glycosylase	Peptidase
	Amidase	Glycosylase	Peptidase
Amidase	3061	9	0
Glycosylase	29	3072	39
Peptidase	0	130	3656

Fig. 2. Projected confusion matrix under GCSV environmental prevalence (30.7% Amidase, 31.4% Glycosylase, 37.9% Peptidase, $n = 10,000$). Glycosylase-Peptidase confusion dominates the off-diagonal entries.

VI. DISCUSSION

The results of this study demonstrate that deep semantic embeddings generated by ESM-2 fundamentally outperform classical biophysical feature engineering for the classification of bacteriophage lysin catalytic domains.

The convergence of Logistic Regression with the complex Multi-Layer Perceptron architecture (yielding $\sim 98.2\%$ baseline accuracy) is a critical insight. It proves that the 150-million-parameter ESM-2 model successfully functions as a structural translator. By projecting raw 1D amino acid sequences into a 640-dimensional latent space, the transformer untangles evolutionary noise and maps the sequences into a topography where the distinct 3D geometries of Amidases,

Glycosylases, and Peptidases become linearly separable. Consequently, the bottleneck in domain classification is no longer the complexity of the downstream classifier, but the quality of the upstream representation.

Furthermore, our rigorous sequence identity analysis validates the absolute necessity of homology-aware data splitting. Because the maximum pairwise sequence identity between the test and training folds averaged 78.89%, a classical random split would have resulted in severe data leakage. By utilizing StratifiedGroupKFold anchored to PhaLP 2.0 cluster identifiers, we ensured that the network’s 98.61% accuracy was achieved by learning the universal biological grammar of the catalytic active sites, rather than memorizing highly identical evolutionary cousins.

Finally, the simulated deployment against the Global Cold Seep Virome (GCSV) [12] confirms the architecture’s viability in real-world deployment. The slight degradation in Glycosylase precision under real-world class imbalances mathematically mirrors the biological reality of multi-domain lysins. Because wild-type lysins frequently fuse distinct enzymatic modules (e.g., an amidase and an endopeptidase) onto a single polypeptide chain to degrade complex peptidoglycan, geometric boundaries inherently overlap in raw environmental samples [2].

VII. CONCLUSION AND FUTURE WORK

This paper introduces PhageLytic-ESM, a pipeline that leverages deep semantic embeddings to bypass the limitations of classical sequence alignment and 1D feature engineering. Evaluated under strict, cluster-aware cross-validation, the tuned MLP architecture achieves 98.61% accuracy in distinguishing Amidase, Glycosylase, and Peptidase catalytic domains, demonstrating an exceptional generalization across isolated evolutionary lineages.

While this study successfully demonstrates high-precision classification using deep semantic embeddings, the current pipeline operates under a closed-world assumption: it assumes the input sequence is already a validated lysin. Future work can focus on developing an upstream Binary Gatekeeper Model. This lightweight, highly efficient pre-filter would rapidly screen raw metagenomic contigs to definitively reject non-lysin sequences. Only validated putative lysins would then be routed to our ESM-2 architecture for final domain classification during large-scale bioprospecting in underexplored extreme environments, such as the Philippine Rise.

Additionally, while the convergence of multiple classifiers around 98% accuracy strongly suggests this represents a near-ceiling for the current dataset and feature representation, it remains possible that larger ESM-2 model variants (e.g., `esm2_t33_650M_UR50D` with 650 million parameters) or fine-tuning approaches could push accuracy further. The current study utilized frozen ESM-2 embeddings as a fixed feature extractor, while end-to-end fine-tuning was not explored due to computational constraints, but it remains a high-value target for future development.

REFERENCES

- [1] V. A. Fischetti, “Bacteriophage lysins as effective antibacterials,” *Current Opinion in Microbiology*, vol. 11, no. 5, pp. 393–400, Oct 2008.
- [2] M. Schmelcher, D. M. Donovan, and M. J. Loessner, “Bacteriophage endolysins as novel antimicrobials,” *Future Microbiology*, vol. 7, no. 10, pp. 1147–1171, Oct 2012.
- [3] S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman, “Basic local alignment search tool,” *Journal of Molecular Biology*, vol. 215, no. 3, pp. 403–410, Oct 1990.
- [4] E. Gasteiger *et al.*, “Protein identification and analysis tools on the expasy server,” in *The Proteomics Protocols Handbook*, J. M. Walker, Ed. Totowa, NJ: Humana Press, 2005, pp. 571–607.
- [5] Z. Lin *et al.*, “Evolutionary-scale prediction of atomic-level protein structure with a language model,” *Science*, vol. 379, no. 6637, pp. 1123–1130, Mar 2023.
- [6] W. Li and A. Godzik, “Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences,” *Bioinformatics*, vol. 22, no. 13, pp. 1658–1659, Jul 2006.
- [7] Y. Park and E. M. Marcotte, “Flaws in evaluation schemes for pair-input computational predictions,” *Bioinformatics*, vol. 28, no. 22, pp. 3058–3065, 2012.
- [8] T. F. Smith and M. S. Waterman, “Identification of common molecular subsequences,” *Journal of Molecular Biology*, vol. 147, no. 1, pp. 195–197, Mar 1981.
- [9] A. Rives *et al.*, “Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences,” *Proceedings of the National Academy of Sciences*, vol. 118, no. 15, p. e2016239118, Apr 2021.
- [10] B. Criel, S. Taelman, W. Van Criekeing, M. Stock, and Y. Briers, “Phalp: A database for the study of phage lytic proteins and their evolution,” *Viruses*, vol. 13, no. 7, p. 1240, Jun 2021.
- [11] A. Boulay, V. Németh, B. Criel, M. Stock, B. De Baets, C. Galiez, E. Rousseau, Y. Briers, and R. Vázquez, “Phalp 2.0: extending the community-oriented phage lysin database with a SUB-LYME pipeline for metagenomic discovery,” *bioRxiv*, Dec 2025, doi: 10.64898/2025.12.08.692814.
- [12] X. Liu, X. Guo, L. Xue, Z. Lu, J. Wang, J. Liao, L. Liu, Y. Chen, Y. Peng, Y. Han, F. Wu, R. Cheng, and X. Dong, “A global atlas of deep-sea cold seep viruses uncovers extensive genomic novelty, pervasive viral dark matter, and biotechnological potential,” *bioRxiv*, Dec 2025, doi: 10.64898/2025.12.10.692286.
- [13] P. J. A. Cock *et al.*, “Biopython: freely available python tools for computational molecular biology and bioinformatics,” *Bioinformatics*, vol. 25, no. 11, pp. 1422–1423, 2009.
- [14] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [15] T. Chen and C. Guestrin, “Xgboost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 785–794.
- [16] B. Rost, “Twilight zone of protein sequence alignments,” *Protein Engineering*, vol. 12, no. 2, pp. 85–94, 1999.